# AdaptLinker at KnowledgeGraphEval 2026: Cross-Domain Arabic Named Entity Recognition with Adversarial and Self-Training

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## Abstract

This paper presents the submission of team **AdaptLinker** to the AdaptNER shared task (Subtask 1: Cross-Domain Named Entity Recognition) at KnowledgeGraphEval 2026. To address nested entities and domain shift across 10 unseen target domains, we propose a multi-head token classification framework built on the AraBERT backbone. Our approach integrates: (1) 21 parallel linear classifiers to predict overlapping named entities; (2) a Domain Adversarial Neural Network (DANN) using a Gradient Reversal Layer (GRL) to align feature spaces; (3) confidence-gated pseudo-labeling on unlabeled target text; and (4) Stochastic Weight Averaging (SWA). Our final system achieved a Precision of 75.48%, Recall of 71.21%, a Micro F1 of 73.29%, and a Macro F1 of 63.37% on the official blinded test set, outperforming the shared-task baseline (0.7329 vs. 0.622 Micro F1) and ranking 6th out of all participating teams. Our code is publicly released on GitHub.[1]

## 1 Introduction

Named Entity Recognition (NER) is essential for Information Extraction and Knowledge Graph Construction. Although supervised models perform well in-domain, performance often degrades markedly on unseen target domains. The AdaptNER shared task (Subtask 1: Cross-Domain Named Entity Recognition) at KnowledgeGraphEval 2026 (Aljabari et al., 2026) (benchmarked alongside WojoodRelations (Aljabari et al., 2025b)) evaluates models on 10 unseen *Konooz* domains (Hamad et al., 2025) (Agriculture, Art, Economics, Finance, Health, History, Law, Politics, Science, Sport) after training on Wojood (Jarrar et al., 2022).

Arabic NER is uniquely challenging due to rich morphology, absence of capitalization, and frequent nested entities where standard flat sequence taggers collapse overlapping mentions. Team **AdaptLinker** addresses these challenges using AraBERT (Antoun et al., 2020) equipped with 21 parallel classification heads, enabling concurrent label predictions per token. To bridge the domain gap, we perform unsupervised adaptation on unlabeled target texts using domain adversarial alignment with a Gradient Reversal Layer (GRL) (Ganin et al., 2016) combined with confidence-gated pseudo-labeling (Lee, 2013; Xie et al., 2020), followed by Stochastic Weight Averaging (SWA) (Izmailov et al., 2018) to seek wider local optima.

## 2 Dataset Analysis & Task Formulation

### 2.1 Dataset Splits & Annotation Scheme

Our supervised source data is the Wojood corpus (Jarrar et al., 2022), annotated with 21 entity categories from WojoodOntology (Aljabari et al., 2025a) spanning general classes (`PERS`, `ORG`, `GPE`) and domain-specific types (`LAW`, `WEBSITE`, `CURR`). Entity nesting is common in Wojood; for example, a geopolitical entity فلسطين (Palestine, `GPE`) is literally contained within an organization mention بنك فلسطين (Bank of Palestine, `ORG`).

Wojood adopts a multi-column CoNLL format where each token receives space-separated BIO tags representing active labels (e.g., فلسطين `I-ORG B-GPE`). Standard flat classification networks collapse overlapping labels and miss nested mentions. Our multi-head architecture preserves this structure by isolating predictions into 21 independent channels. The Wojood training corpus exhibits an extreme long-tailed label distribution: frequent categories such as `GPE` (2,156 validation mentions) and `ORG` (1,882 mentions) dominate, whereas minority classes like `PRODUCT` (8 mentions) and `QUANTITY` (3 mentions) provide sparse

[1]https://github.com/ArgoHyperion/Multi-Head-AraBERT-with-Domain-Adversarial-Training-and-Self-Training-for-Arabic-NER



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supervision.

The target domain is represented by the *Konooz* corpus (Hamad et al., 2025), which spans 10 specialized domains. To prevent data leakage, splits are strictly isolated: supervised pre-training uses the official Wojood training set; in-domain validation is tracked on the Wojood validation set; Phase 2 adaptation strictly uses the unlabeled *Konooz* development set across the 10 domains; and the blinded *Konooz* test set (1,645 sentences) was held out exclusively for final competition evaluation.

### 2.2 Task Formulation

Given token representations $H \in \mathbb{R}^{L\times 768}$ from the AraBERT backbone, we define 21 parallel linear classifiers, one for each entity type:

$$\hat{y}_t^{(c)} = \text{Softmax}(W^{(c)} h_t + b^{(c)}) \quad (1)$$

where $W^{(c)} \in \mathbb{R}^{3\times 768}$ and $b^{(c)} \in \mathbb{R}^3$ parameterize the classifier for category $c$, and $\hat{y}_t^{(c)} \in [0,1]^3$ is the predicted BIO distribution $\{B, I, O\}$ for token $t$ and entity type $c$. By predicting each entity channel independently, the model represents tokens belonging to multiple entity types simultaneously.

Because each entity head outputs a single BIO tag per token, this formulation supports arbitrary overlapping mentions across different entity types, but cannot natively represent concentric spans sharing the identical entity label (e.g., an `ORG` strictly nested inside another `ORG`). In Wojood, such same-type nested mentions do not occur in the annotation schema, making the 21-head factorization well-suited to the task.

## 3 System Methodology

Our pipeline operates in two sequential phases: supervised source pre-training (Phase 1) and unsupervised target adaptation (Phase 2), followed by SWA checkpoint aggregation.

### 3.1 Phase 1: Supervised Pre-training

In Phase 1, the network is trained on the labeled Wojood training set by minimizing the summed multi-head cross-entropy loss:

$$\mathcal{L}_{\text{ner_src}} = \sum_{c=1}^{21} \mathcal{L}_{\text{CE}}(\hat{y}^{(c)}, y^{(c)}) \quad (2)$$

Subword alignment assigns the gold token BIO labels to the first subword piece of each word, while setting subsequent subwords to `ignore_index=-100`. This standard masking ensures that loss computation is evaluated exactly once per word token, preventing multi-subword tokens from exerting disproportionate gradient weight.

### 3.2 Phase 2: Unsupervised Domain Adaptation

In Phase 2, the model adapts to *Konooz* using labeled source batches (Wojood training set) and unlabeled target batches (*Konooz* development set).

**Domain Adversarial Alignment** We align the latent feature distributions of the source and target domains via a domain classifier operating on the pooled `[CLS]` representation. The domain head is a two-layer MLP: $\text{Linear}(768, 256) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.2) \rightarrow \text{Linear}(256, 2)$ (Srivastava et al., 2014). A Gradient Reversal Layer (GRL) (Ganin et al., 2016) is inserted between the backbone and the domain classifier. In the forward pass, GRL acts as an identity operator; in the backward pass, it multiplies incoming gradients by $-\lambda$, encouraging the encoder to learn domain-invariant representations: $\lambda = 2/(1 + e^{-10\cdot p}) - 1$, where $p = \text{step}/\text{total_steps}$. The domain classification loss is $\mathcal{L}_{\text{domain}} = \mathcal{L}_{\text{CE}}(\hat{d}, d)$, where $d \in \{0, 1\}$ denotes source vs. target domain. The binary source-vs-target discriminator provides a domain-adversarial objective for reducing aggregate source-target representation discrepancy (Ganin et al., 2016) without requiring fine-grained domain partitions or target labels.

**Confidence-Gated Pseudo-Labeling** To transfer task knowledge to the target domain, we employ a one-shot teacher-student framework with a fixed teacher. At the start of Phase 2, a frozen copy of the best Phase 1 checkpoint is instantiated as the teacher model $\theta_{\text{teach}}$, while the student model $\theta_{\text{stud}}$ is initialized from the same Phase 1 weights (with its bottom 6 transformer layers frozen). For target sentences, the teacher generates predictions with gradients disabled. We apply token-level confidence gating independently to each token $t$ and entity type $c$:

$$\mathcal{L}_{\text{st_tgt}} = \sum_{c=1}^{21} \frac{1}{|S_c|} \sum_{t\in S_c} \mathcal{L}_{\text{CE}}(\hat{y}_{t,\text{stud}}^{(c)}, y_{t,\text{pseudo}}^{(c)}) \quad (3)$$

where $S_c = \{t \mid \max_k \hat{y}_{t,\text{teach}}^{(c)}(k) > \tau\}$ is the subset of tokens whose maximum teacher probability for class $c$ exceeds the threshold $\tau = 0.90$, and

$y^{(c)}_{t,\text{pseudo}}$ is the corresponding predicted tag. Filtering at the token $\times$ entity-type level ensures that only highly confident predictions provide pseudo-supervision, while ambiguous tokens are excluded without discarding entire sentences. The teacher remains frozen throughout Phase 2 to prevent confirmation bias drift.

The student model and domain classifier are jointly optimized by minimizing $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ner_src}} + 0.8\mathcal{L}_{\text{st_tgt}} + 0.3\mathcal{L}_{\text{domain}}$. The self-training weight (0.8) down-weights pseudo-labeled gradients relative to ground-truth Wojood labels to curb noise propagation, while the conservative domain weight (0.3) avoids distorting token representations (Ganin et al., 2016; Yadav and Bethard, 2018).

### 3.3 Stochastic Weight Averaging (SWA)

Following adaptation, we average parameter weights across the final $M = 4$ adaptation checkpoints:

$$\theta_{\text{SWA}} = \frac{1}{M} \sum_{m=1}^{M} \theta_m \tag{4}$$

Because AraBERT uses Layer Normalization rather than Batch Normalization, uniform weight averaging does not require recomputing running statistics. SWA is motivated as a regularizer to encourage convergence toward flatter loss basins that can favor generalization (Izmailov et al., 2018).

## 4 Experimental Setup

Experiments were implemented in PyTorch using Hugging Face transformers on Kaggle Cloud (2$\times$ NVIDIA Tesla T4 GPUs, 30GB RAM, random seed 42).

**Preprocessing & Backbone:** Sequences are tokenized via AraBERT's WordPiece tokenizer (`bert-base-arabertv02`) with maximum length 512 and first-subword alignment.

**Phase 1 (Supervised):** Trained using AdamW (Loshchilov and Hutter, 2017) (learning rate $2 \times 10^{-5}$ (Devlin et al., 2019), ExponentialLR $\gamma = 0.95$, batch size 8, dropout 0.1 (Srivastava et al., 2014)) for up to 50 epochs. Early stopping (patience 5) on Wojood validation Micro F1 terminated training at Epoch 7 (8 epochs total).

**Phase 2 (Adaptation):** Trained for 5 epochs on the combined Wojood training set and unlabeled *Konooz* development set (batch size 4, weight decay 0.01 (Krogh and Hertz, 1991), learning rate $2 \times 10^{-5}$ (Howard and Ruder, 2018; Sun et al., 2019)). The bottom 6 layers of AraBERT (embeddings + layers 0–5) are frozen following transfer learning findings that lower transformer layers capture general linguistic features (Kovaleva et al., 2019; Lee et al., 2020), while layers 6–11 and classification heads adapt. SWA is applied across the last 4 checkpoints. Hyperparameters ($\tau = 0.90$, loss weights 0.8 and 0.3, 6 frozen layers, 5 epochs, SWA over 4 checkpoints) were selected during development to balance target adaptation against source retention, drawing on transfer learning conventions for BERT backbones (Kovaleva et al., 2019; Howard and Ruder, 2018).

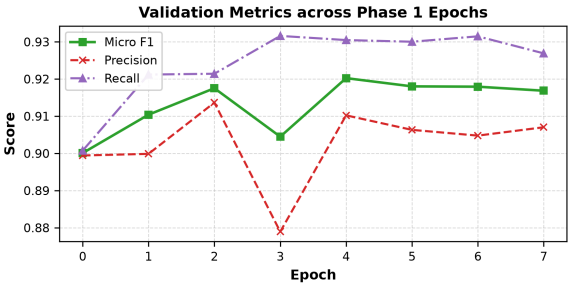



Figure 1: Validation Micro F1, Precision, and Recall metrics over Phase 1 (supervised training) epochs on the Wojood validation set (Seed 42).

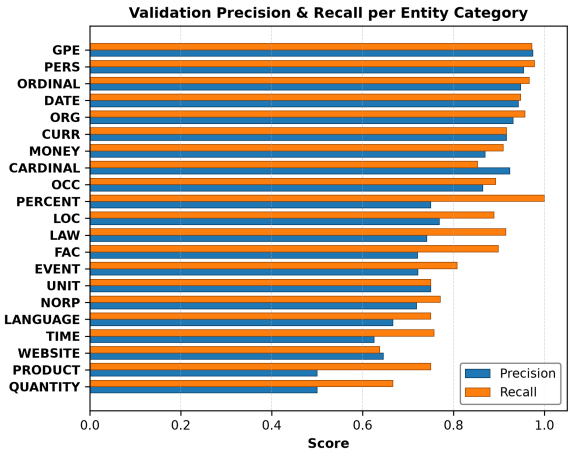



Figure 2: Per-category validation Precision and Recall on the Wojood validation set (evaluated on the post-adaptation SWA model).

## 5 Results & Discussion

### 5.1 In-Domain Validation Performance

Figure 1 tracks Phase 1 training. Validation loss reaches its minimum at Epoch 2 (0.0874, F1: 0.9175), whereas validation Micro F1 peaks at Epoch 4 (0.9202, Precision: 91.02%, Recall: 93.05%). This illustrates that validation loss can

| System (Evaluation Split) | Precision | Recall | Micro F1 |
|---|---|---|---|
| AdaptLinker (Wojood Val) | 0.9089 | 0.9330 | 0.9208 |
| Shared Baseline (*Konooz* Test) | 0.6300 | 0.4400 | 0.6220 |
| AdaptLinker (*Konooz* Test) | **0.7548** | **0.7121** | **0.7329** |

Table 1: System performance on Wojood validation and official *Konooz* test sets alongside the baseline. AdaptLinker Macro F1: 0.8151 (val), 0.6337 (test).

decouple from entity F1 under 'O'-class dominance, supporting validation Micro F1 as the early-stopping metric.

Phase 2 adaptation initialized from this Epoch 4 model; at adaptation epoch 0, the model achieved 0.9211 Micro F1 on the Wojood validation set. Averaging the final 4 checkpoints via SWA yielded the final model $\theta_{\text{SWA}}$, scoring 0.9208 Micro F1 (0.9089 Precision, 0.9330 Recall, 0.8151 Macro F1) on the Wojood validation set (Table 1).

Figure 2 details class-wise Precision and Recall for $\theta_{\text{SWA}}$. The system achieves high performance on dominant categories such as `GPE` (0.9735 F1) and `PERS` (0.9658 F1), but exhibits lower scores on rare classes like `PRODUCT` (0.6000 F1) and `QUANTITY` (0.5714 F1), consistent with their sparse representation in the data.

### 5.2 Official Test Evaluation

Table 1 reports official competition results on the blinded test set (1,645 sentences across 10 *Konooz* domains). The final SWA model achieved an official Micro F1 of 0.7329 (0.7548 Precision, 0.7121 Recall, 0.6337 Macro F1), placing 6th, compared with 0.622 Micro F1 (0.63 Precision, 0.44 Recall) for the baseline.

The 18.79% drop in Micro F1 illustrates cross-domain difficulty. Higher Precision (75.48%) than Recall (71.21%) may reflect more conservative predictions out-of-domain. We hypothesize that while adversarial alignment and pseudo-labeling capture shared structural features, pronounced lexical and syntactic shifts disproportionately suppress recall on domain-specific entities. Moreover, the Micro F1 (0.7329) vs. Macro F1 (0.6337) gap is consistent with the long-tailed class distribution: frequent categories transfer more reliably, whereas rare categories exhibit lower scores, lowering the unweighted macro average.

## 6 Error Analysis

**Out-of-Vocabulary Subword Fragmentation:** In technical domains (Health, Science, Law), specialized Arabic terminology is infrequent in AraBERT's vocabulary. The tokenizer fractures these terms into multiple subwords; we hypothesize this weakens span representations, contributing to missed entities and lower recall (71.21%).

**Boundary Ambiguity in Nested Mentions:** Because the 21 classification heads operate independently without transition decoding constraints, overlapping predictions can exhibit boundary inconsistencies where one head detects a span but an overlapping head misses or truncates it.

**Minority-Class Pseudo-Label Sparsity:** We hypothesize that a fixed threshold ($\tau = 0.90$) disadvantages low-support categories (e.g., `PRODUCT` with 8 validation instances, `QUANTITY` with 3). Because the teacher rarely predicts rare entities with high confidence, the student receives negligible pseudo-supervision for minority classes on target text.

## 7 Limitations

**Domain Shift Sensitivity:** Unsupervised alignment using GRL and pseudo-labeling cannot fully bridge pronounced lexical shifts in technical domains like Law and Medicine.

**Pseudo-Label Noise:** Confident but erroneous teacher predictions introduce noisy supervision, which can propagate confirmation bias to the student model.

**Structural Representation Constraint:** Factorizing the label space into 21 independent heads cannot represent concentric nested mentions of the identical entity type, though this constraint matches Wojood's annotation schema.

**Computational Footprint:** Dual forward passes (source and target batches) and maintaining a separate teacher model increase memory requirements during adaptation.

## 8 Conclusion & Future Work

We presented AdaptLinker, a multi-head AraBERT framework for cross-domain Arabic nested NER combining domain adversarial training, pseudo-labeling, and SWA. AdaptLinker achieved an official Micro F1 of 0.7329 on the out-of-domain *Konooz* test set (vs. 0.622 for the baseline), placing 6th. Future work includes character encoders (CANINE), multi-domain objectives, and component ablations.

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